

Can Concept Erasure Reproduce Concept Exclusion? A Matched Evaluation of EMBER, RMU, and SNMF

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Study design schematic to draw from final protocol

Figure 1: Planned comparison of the full model, concept-excluded twin, and three erasure methods on identical target, neighboring, and unrelated sentence completions.

Abstract

Plan: Target 150–190 words. State the erasure question; matched full/twin/ EMBER/RMU/SNMF design; three concepts and three evaluation groups; answer NLL and full-vocabulary KL; the most important result and qualification once known. Write last. No numbers until all five models use the same held-out protocol.

1 Introduction

Plan: About 0.7 page. Motivate selective concept erasure and explain why “forgetting” needs two checks: approaching the concept-excluded twin on the target and retaining correct-answer support elsewhere. Define the full model, excluded twin, and post-hoc edit before using those terms repeatedly. Name Ancient Rome, Baseball, and AI. State that the study measures answer-token probability and next-token distributional similarity, not generated-answer accuracy. End with 2–3 concrete contributions once results exist.

2 Background and Related Work

2.1 LMEnt and Concept Exclusion

LMEnt provides a Wikipedia-based pretraining corpus annotated with entity mentions, an index for retrieving chunks associated with those entities, and language models trained on the corpus (Gottesman et al., 2025). The entity index lets us identify training chunks linked to a target concept. By masking the loss on those chunks during an otherwise matched training run, we obtain a *concept-excluded*

twin. This twin provides a reference for the effect of excluding a defined portion of the training data. It may still learn related facts from chunks outside that portion.

2.2 Post-Training Concept Erasure

EMBER edits token embeddings rather than internal transformer layers (Suslik et al., 2026). It factorizes the embedding matrix into sparse features, selects features associated with the target concept, and subtracts their contributions from the embeddings of selected tokens. This makes it possible to change a small set of input representations without updating the rest of the model.

We also consider two approaches that act on multilayer perceptron (MLP) components. Representation Misdirection for Unlearning (RMU) updates selected MLP parameters to redirect representations of forget-set inputs, while a retain objective encourages representations of other inputs to remain close to the original model (Li et al., 2024). Semi-nonnegative matrix factorization (SNMF) identifies interpretable features in MLP activations from groups of co-activated neurons (Shafran et al., 2026). In its erasure adaptation, concept-associated features are mapped to directions that can be removed through MLP weight edits (Suslik et al., 2026).

The EMBER study evaluated erasure using multiple-choice and open-ended questions about target concepts, with related-topic questions and general benchmarks used to assess preservation (Suslik et al., 2026). It also examined generation quality and whether target knowledge returned after relearning. These evaluations compared edited models with the original model, which had been trained on the target concept. They included EMBER, RMU, and an MLP weight-erasure adaptation of SNMF-derived features (Li et al., 2024; Shafran et al., 2026). RMU’s original evaluation likewise measured target-question performance and

general capabilities relative to the original model. Our study adds a matched concept-excluded twin as a reference and compares edited models with it using correct-answer NLL and full-vocabulary KL on the same sentence completions.

3 Experimental Design

3.1 Models and Matched Training

We trained OLMo-2 1B models on LMEnt’s entity-annotated Wikipedia corpus (Team OLMo et al., 2025; Gottesman et al., 2025). The shared control model received loss on the full corpus. For each target concept, we trained a concept-excluded twin from the same initialization, using the same batch order, optimizer, learning-rate schedule, and two-epoch duration. Exclusion was applied after batch construction: labels in selected chunks were masked so that those chunks contributed no training loss. Each twin therefore differed from the shared control in the learning signal from its selected chunks, while the training procedure remained matched.

3.2 Concepts and Exclusion Sets

We selected Ancient Rome, Baseball, and artificial intelligence (AI) by hand before training the erasure models. Preliminary accuracy of the full model on target selection questions helped us avoid topics with little measurable knowledge. We also sought topics of manageable breadth and variation across domains. Because this was a purposive choice, our findings don’t necessarily generalize to every concept.

For each concept, we used Claude in a dedicated session to propose Wikidata entity IDs (QIDs), then reviewed their coverage in LMEnt’s corpus (Gottesman et al., 2025). A single ID missed text linked to related entities, while an overly broad set risked excluding text about adjacent topics. We therefore selected a bounded set of QIDs for each concept. We excluded a training chunk if LMEnt linked at least one mention in it to a selected QID. Table 1 reports the resulting ID and unique chunk counts.

Concept	Entity IDs	Excluded chunks
Ancient Rome	56	65,844
Baseball	43	80,466
AI	31	19,818

Table 1: Entity sets used to exclude concept-associated training chunks.

3.3 Post-Training Erasure and Checkpoint Selection

We applied EMBER, RMU, and an SNMF-based erasure independently to copies of the completed control model for each topic. EMBER edited selected input embeddings, RMU updated selected MLP parameters, and the SNMF-based method removed concept-associated directions through MLP weight edits. Each topic received the same method-specific procedure and checkpoint-selection rule.

We selected one checkpoint per method and topic using answer NLL on selection questions. Let $\Delta_T(M)$ be candidate M ’s mean NLL minus the control’s on target questions. For neighboring and unrelated questions, let $D_g(M)$ be the mean absolute per-question NLL difference between M and the control. We selected the candidate maximizing

$$S(M) = 0.5\Delta_T(M) - 0.25D_{\text{neighbor}}(M) - 0.25D_{\text{unrelated}}(M). \quad (1)$$

The unrelated selection set contained 50 questions sampled from topics outside the target and neighboring domains; a separate 50-question set was reserved for testing. The concept-excluded twins and all test questions were excluded from checkpoint selection. We subsequently compared each selected erasure checkpoint with both the control and its concept-excluded twin using the protocol in Section 4.

4 Evaluation Protocol

4.1 Three Groups and Fixed Correct Answers

For each concept, we evaluated target questions, neighboring-topic questions, and questions from unrelated topics. The target group measures change on the concept to be erased; the other groups assess preservation at different degrees of topical proximity. Each group contained 50 selection and 50 test questions. The target and neighboring questions followed EMBER’s existing splits (Suslik et al., 2026); we sampled the unrelated questions from other topics.

We used Claude in a dedicated session to rewrite the multiple-choice questions as declarative sentence stems while retaining their correct answers. For example, “Which river is associated with the founding of Rome?” became “The river associated with the founding of Rome is”, with “The Tiber

River” as the correct continuation. This format avoided relying on the 1B base models to produce a gradeable response to a multiple-choice instruction, which they often failed to do in preliminary tests.

Ranking the supplied options would give disproportionate weight to three particular wrong answers, although many other incorrect continuations are possible. We therefore scored only the correct continuation.

4.2 Answer NLL Difference

For question i , let x_i be the sentence stem and $y_{i,1}, \dots, y_{i,n_i}$ the tokens of its correct continuation. We measured model M ’s mean negative log-likelihood (NLL) over those answer tokens:

$$\ell_M(i) = -\frac{1}{n_i} \sum_{t=1}^{n_i} \log p_M(y_{i,t} \mid x_i, y_{i,<t}). \quad (2)$$

Each token was scored given the stem and the preceding correct-answer tokens; the model did not generate a response.

To compare M with reference model R , either the full model or the concept-excluded twin, we computed the paired difference $\Delta_{M,R}(i) = \ell_M(i) - \ell_R(i)$. A positive difference means that M assigned less probability to the correct answer than R .

4.3 Teacher-Forced Full-Vocabulary KL

Answer NLL measures support for the supplied answer. To measure broader changes in prediction, we also compared the models’ full next-token distributions at each answer position. Let $p_{M,t}$ denote model M ’s distribution over the vocabulary given the stem and the preceding correct-answer tokens. For concept-excluded twin T and comparison model M , we computed

$$\text{KL}_{T \rightarrow M}(i) = \frac{1}{n_i} \sum_{t=1}^{n_i} D_{\text{KL}}(p_{T,t} \parallel p_{M,t}). \quad (3)$$

We kept the twin on the left when comparing it with both the full model and each erased model. This choice matters because KL is asymmetric: reversing its arguments can change the values and even the ordering of models. Using the twin as the common reference asks how closely each model reproduces the twin’s predictions, with deviations weighted by the twin’s probability distribution. Lower KL indicates greater similarity at the evaluated positions.

Signed answer-NLL changes by concept, model, and question group

Figure 2: Planned answer-NLL differences from the full model on target, neighboring, and unrelated sentence completions, with paired intervals.

The full vocabulary includes incorrect next tokens, so KL can detect changes among alternatives even when the correct token’s probability changes little. Because the models are evaluated along the correct continuation, KL does not measure their predictions after an incorrect token is chosen.

We report the mean and standard deviation across the 50 held-out test questions in each target, neighboring, and unrelated group. For NLL, these summarize paired question-level differences; for KL, they summarize question-level divergence.

5 Results

5.1 What Concept Exclusion Changes

Plan: About 0.4 page. For each concept, establish twin-minus-full answer NLL and $\text{KL}(\text{full} \rightarrow \text{twin})$ on target questions, then neighboring and unrelated questions. State where the exclusion reference has weak observable effect. Point to complete per-group twin rows in Appendix D.

5.2 How Closely Erasure Matches the Twin on Target Items

Plan: About 0.7 page. Compare EMBER, RMU and SNMF with the twin and full model on target answer NLL and full-vocabulary KL. Show paired intervals and inspect per-question variation; a near-zero signed mean NLL does not establish identical behavior. Discuss substantive differences among Rome, Baseball, and AI only after the new measurements are filled.

5.3 Neighboring and Unrelated Correct-Answer Support

Plan: About 0.6 page. Give neighboring and unrelated answer-NLL and KL comparisons separately for every method. Positive NLL from full indicates weaker prediction of the correct answer; KL indicates a distributional change but has no direction of quality. Discuss neighboring spillover and unrelated preservation relative to both full and twin.

5.4 Where Answer NLL and KL Disagree

Plan: About 0.3 page. Examine cases where the correct-answer probability appears stable but the full next-token dis-

Concept	Erasure model	Δ NLL vs full	Δ NLL vs twin	KL(twin $\rightarrow$ erasure)
Rome	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Baseball	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AI	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 2: Target-question summary to populate after common-protocol scoring. NLL changes are signed evaluated-minus-reference differences. Show paired intervals in the final version and the twin-minus-full reference effect nearby.

tribution changes, or vice versa. Check whether a near-zero mean signed NLL difference hides opposing question-level changes. Use concrete paired examples only if supported by the final data.

6 Analysis and Discussion

6.1 What Counts as Matching Concept Exclusion?

Plan: About 0.35 page. Synthesize target similarity and preservation across answer NLL and KL. Explain why higher target answer NLL than the twin does not establish a better edit, and why low KL alone cannot establish knowledge preservation. Respect the scope of fixed-answer sentence completions.

6.2 Why Concepts and Metrics May Disagree

Plan: About 0.45 page. Investigate supported explanations: blacklist coverage, neighboring-topic overlap, per-question NLL cancellation, and disagreements between correct-answer NLL and full-vocabulary KL. Clearly distinguish observed evidence from hypotheses. Add a small diagnostic figure only if it clarifies a central result; otherwise put diagnostics in Appendix E.

7 Limitations

Plan: About 0.4 page. State limits from operational blacklist coverage, fixed correct answers rather than generated-answer accuracy, no general passage-prediction measurement, single training seed/checkpoint per arm, finite and potentially dependent questions, and teacher-forced KL. Identify any exploratory selection or unequal evaluation cell honestly when results are filled.

8 Conclusion

Plan: About 0.2 page. Answer which method most closely matches each twin on correct-answer NLL and full-vocabulary KL, and whether neighboring/unrelated answer support is preserved, at the level actually supported. Do not claim

generated-answer accuracy or passage-level erasure. One restrained next step.

AI Disclosure and Reflection

Plan: About 0.2 page. Fill with the actual AI tools and models the team used, their roles in coding, editing, or analysis, and a concise reflection on verification. The course guidelines require specific names; do not guess.

References

- Daniela Gottesman, Alon Gilae-Dotan, Ido Cohen, Yoav Gur-Arieh, Marius Mosbach, Ori Yoran, and Mor Geva. 2025. LMEnt: A suite for analyzing knowledge in language models from pretraining data to representations. *Preprint*, arXiv:2509.03405.
- Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, Justin D. Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, Gabriel Mukobi, Nathan Helm-Burger, Rassin Lababidi, Lennart Justen, Andrew B. Liu, Michael Chen, Isabelle Barrass, Oliver Zhang, Xiaoyuan Zhu, et al. 2024. The WMDP benchmark: Measuring and reducing malicious use with unlearning. In *Proceedings of the 41st International Conference on Machine Learning*, pages 28525–28550.
- Or David Shafran, Atticus Geiger, and Mor Geva. 2026. Constructing interpretable features from compositional neuron groups. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 42326–42348.
- Clara Haya Suslik, Or Shafran, and Mor Geva. 2026. Don’t forget your embeddings: Robust knowledge erasure via precise editing of embeddings. *Preprint*, arXiv:2606.03695.
- Team OLMO, Pete Walsh, Luca Soldaini, Dirk Groeneweld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark,

Pradeep Dasigi, Nouha Dziri, and 21 others. 2025. 2
OLMo 2 furious. *Preprint*, arXiv:2501.00656.

A Corpus Exclusion and Model Configurations

Plan: Table A1: one row per concept and method with checkpoint ID, erasure settings, training/selection data, edited parameters, and seed. Give exact blacklist rules and audit estimates. Do not paste raw run logs.

B Sentence-Completion Evaluation Sets

Plan: Table A2: target, neighboring, and unrelated question counts for each concept; item construction and exclusions; one representative fixed prompt/answer; question provenance and train/selection/test overlap checks.

C Scoring and Statistical Details

Plan: Write exact equations for per-question answer-token NLL, signed evaluated-minus-reference group mean, and full-vocabulary $KL(\text{reference} \rightarrow \text{evaluated})$ under correct-answer teacher forcing. Specify tokenization, prefix, answer-token mask, position/question aggregation, bootstrap resampling and paraphrase clustering. Include one worked item if useful. No option-accuracy or passage-NLL metrics belong in this protocol.

D Complete Results by Concept

D.1 Ancient Rome

Plan: Table A3: all five models, all three question groups, answer NLL and full-vocabulary KL; underlying group means, full/twin contrasts, and paired intervals. Split by metric if needed.

D.2 Baseball

Plan: Table A4 with the identical structure and question-level protocol.

D.3 Artificial Intelligence

Plan: Table A5 with the identical structure and question-level protocol.

E Diagnostic Analyses

Plan: Optional: per-question signed answer-NLL differences showing possible cancellation; examples where NLL and KL disagree; distributional plots and supported case studies. Include only analyses actually performed and needed to interpret main results.

F Reproducibility Notes

Plan: Final scoring script/version, model IDs, seeds, exact question manifest and split chronology, tokenization/context settings, answer scoring masks, and any deviations from the common protocol. Keep this concise.